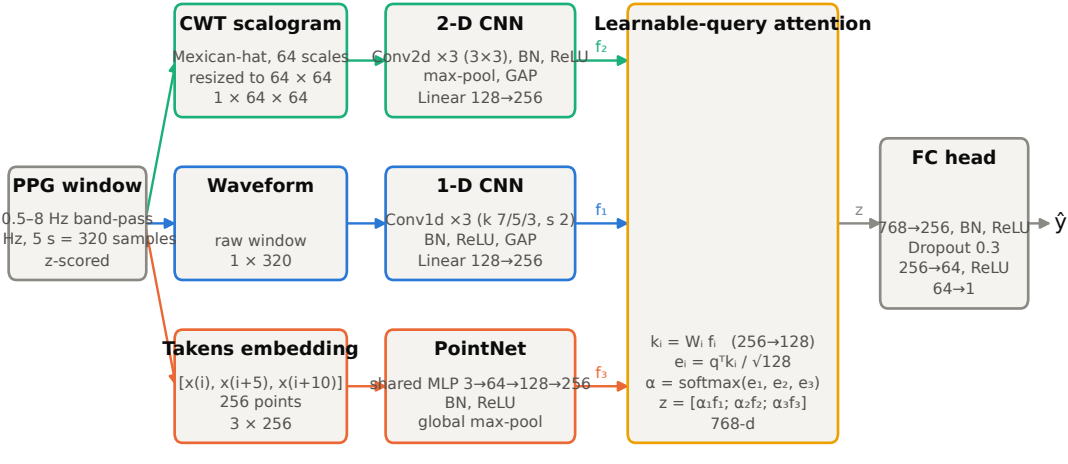
PPG window
0.5–8 Hz band-pass
64 Hz, 5 s = 320 samples
z-scored
CWT scalogram
Mexican-hat, 64 scales
resized to 64 × 64
1 × 64 × 64
2-D CNN
Conv2d ×3 (3×3), BN, ReLU
max-pool, GAP
Linear 128→256
f₂
Waveform
raw window
1 × 320
1-D CNN
Conv1d ×3 (k 7/5/3, s 2)
BN, ReLU, GAP
Linear 128→256
f₁
Takens embedding
[x(i), x(i+5), x(i+10)]
256 points
3 × 256
PointNet
shared MLP 3→64→128→256
BN, ReLU
global max-pool
f₃
Learnable-query attention
kᵢ = Wᵢ fᵢ (256→128)
eᵢ = qᵀkᵢ / √128
α = softmax(e₁, e₂, e₃)
z = [α₁f₁; α₂f₂; α₃f₃]
768-d
z
FC head
768→256, BN, ReLU
Dropout 0.3
256→64, ReLU
64→1
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